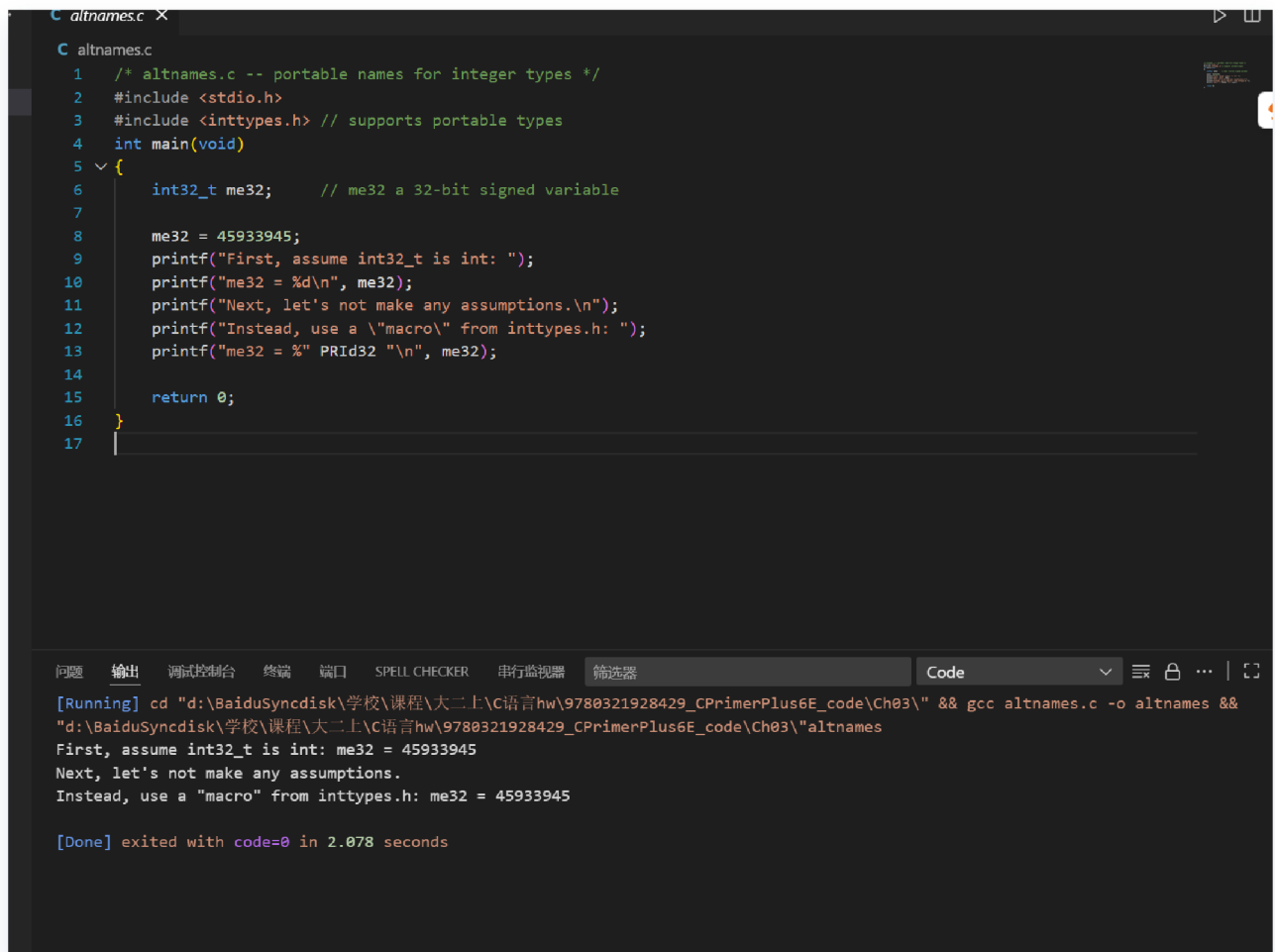


第1章预习报告

altnames.c

- 作用: 演示可移植整数类型的使用
- 功能: 展示如何使用inttypes.h头文件中的int32_t类型和PRId32宏来确保代码在不同平台上的可移植性
- 演示效果:



```
altnames.c
1  /* altnames.c -- portable names for integer types */
2  #include <stdio.h>
3  #include <inttypes.h> // supports portable types
4  int main(void)
5  {
6      int32_t me32;      // me32 a 32-bit signed variable
7
8      me32 = 45933945;
9      printf("First, assume int32_t is int: ");
10     printf("me32 = %d\n", me32);
11     printf("Next, let's not make any assumptions.\n");
12     printf("Instead, use a \"macro\" from inttypes.h: ");
13     printf("me32 = %\" PRId32 \"\n", me32);
14
15     return 0;
16 }
17
```

问题 输出 调试控制台 终端 端口 SPELL CHECKER 串行监视器 筛选器 Code

```
[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\" && gcc altnames.c -o altnames &&
"d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\"altnames
First, assume int32_t is int: me32 = 45933945
Next, let's not make any assumptions.
Instead, use a "macro" from inttypes.h: me32 = 45933945

[Done] exited with code=0 in 2.078 seconds
```

badcount.c

- 作用: 演示printf函数参数数量错误
- 功能: 展示三种常见的printf参数错误: 参数过多、参数过少、参数类型不匹配
- 语义错误:

第9行: `printf("%d\n", n, m);` - 格式字符串只有1个%d, 但提供了2个参数

第10行: `printf("%d %d %d\n", n);` - 格式字符串有3个%d, 但只提供了1个参数

第11行: `printf("%d %d\n", f, g);` - 格式字符串需要整数, 但提供了浮点数

- 演示效果:

修复前:

```
C badcount.c
1  /* badcount.c -- incorrect argument counts */
2  #include <stdio.h>
3  int main(void)
4  {
5      int n = 4;
6      int m = 5;
7      float f = 7.0f;
8      float g = 8.0f;
9
10     printf("%d\n", n, m);    /* too many arguments */
11     printf("%d %d %d\n", n); /* too few arguments */
12     printf("%d %d\n", f, g); /* wrong kind of values */
13
14     return 0;
15 }
16
```

问题 输出 调试控制台 终端 端口 SPELL CHECKER 串行监视器 筛选器 Code

First, assume `int32_t` is `int`: `me32 = 45933945`
Next, let's not make any assumptions.
Instead, use a "macro" from `inttypes.h`: `me32 = 45933945`

[Done] exited with code=0 in 2.078 seconds

[Running] `cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\" && gcc badcount.c -o badcount && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\"badcount`

4
4 5 6422280
0 1075576832

[Done] exited with code=0 in 0.302 seconds

修复后:

```
C badcount.c X
C badcount.c
1  /* badcount.c -- incorrect argument counts */
2  #include <stdio.h>
3  int main(void)
4  {
5      int n = 4;
6      int m = 5;
7      float f = 7.0f;
8      float g = 8.0f;
9
10     printf("%d %d\n", n, m); /* too many arguments */
11     printf("%d\n", n); /* too few arguments */
12     printf("%f %f\n", f, g); /* wrong kind of values */
13
14     return 0;
15 }
16

问题 输出 调试控制台 终端 端口 SPELL CHECKER 串行监视器 筛选器 Code
4
4 5 6422280
0 1075576832

[Done] exited with code=0 in 0.302 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\" && gcc badcount.c -o badcount &&
"d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\"badcount
4 5
4
7.000000 8.000000

[Done] exited with code=0 in 0.314 seconds
```

bases.c

- **作用:** 演示不同进制数的输出
- **功能:** 展示如何使用printf输出十进制、八进制和十六进制数，以及使用#标志显示进制前缀
- **演示效果:**

```
C badcount.c  C bases.c  X
C bases.c
1  /* bases.c--prints 100 in decimal, octal, and hex */
2  #include <stdio.h>
3  int main(void)
4  {
5      int x = 100;
6
7      printf("dec = %d; octal = %o; hex = %x\n", x, x, x);
8      printf("dec = %d; octal = %o; hex = %#x\n", x, x, x);
9
10     return 0;
11 }
12

问题 输出 调试控制台 终端 端口 SPELL CHECKER 串行监视器 筛选器 Code
"d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\"badcount
4 5
4
7.000000 8.000000

[Done] exited with code=0 in 0.314 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\" && gcc bases.c -o bases &&
"d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\"bases
dec = 100; octal = 144; hex = 64
dec = 100; octal = 0144; hex = 0x64

[Done] exited with code=0 in 0.305 seconds
```

charcode.c

- **作用:** 显示字符的ASCII码值
- **功能:** 读取用户输入的字符，然后显示该字符及其对应的ASCII码值
- **演示效果:**

```
charcode.c
1  /* charcode.c-displays code number for a character */
2  #include <stdio.h>
3  int main(void)
4  {
5      char ch;
6
7      printf("Please enter a character.\n");
8      scanf("%c", &ch); /* user inputs character */
9      printf("The code for %c is %d.\n", ch, ch);
10
11     return 0;
12 }
13
```

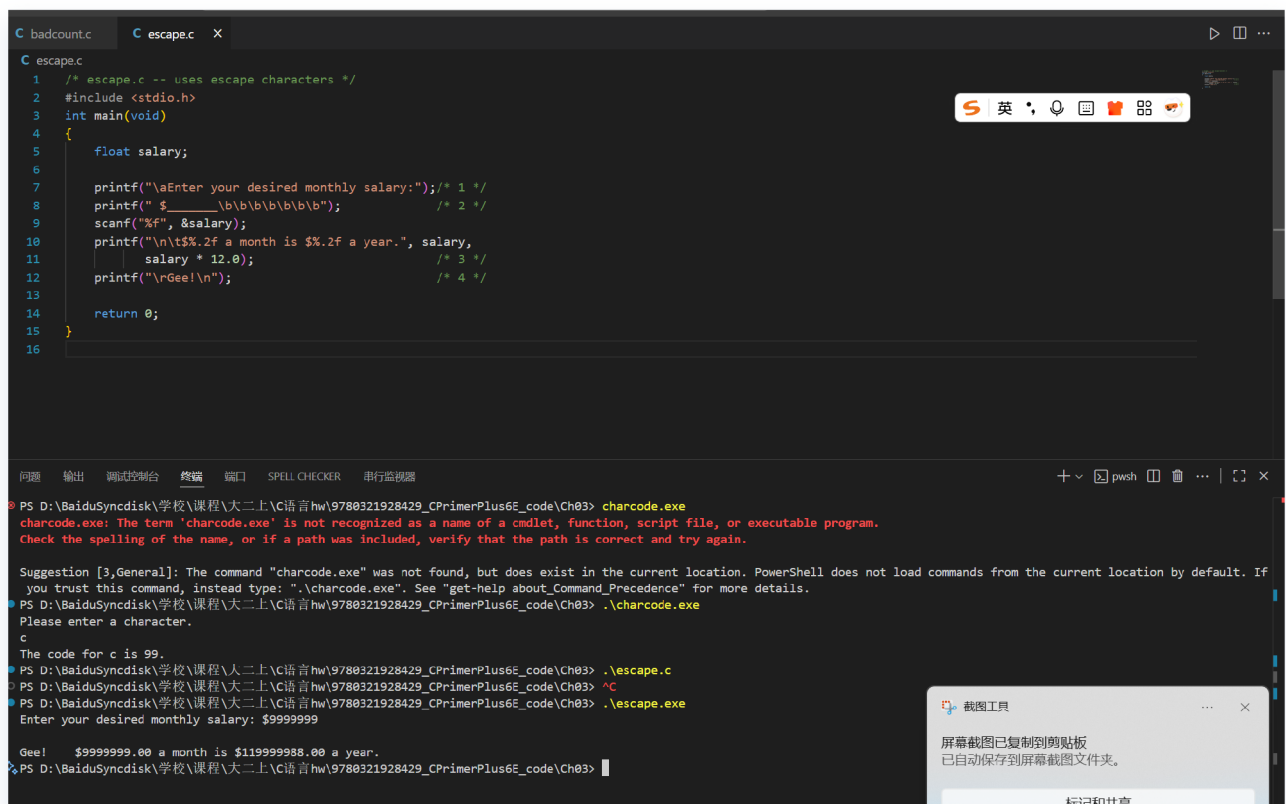
问题 输出 调试控制台 终端 窗口 SPELL CHECKER 串行监视器

```
PS D:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03> charcode.exe
charcode.exe: The term 'charcode.exe' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

Suggestion [3,General]: The command "charcode.exe" was not found, but does exist in the current location. PowerShell does not load commands from the current location by default. If
you trust this command, instead type: ".\charcode.exe". See "get-help about_Command_Precedence" for more details.
PS D:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03> .\charcode.exe
Please enter a character.
c
The code for c is 99.
PS D:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03>
```

escape.c

- **作用:** 演示转义字符的使用
- **功能:** 展示[\a](#)（响铃）、[\b](#)（退格）、[\t](#)（制表符）、[\r](#)（回车）等转义字符的效果
- **演示效果:**



floater.c

- **作用:** 演示浮点数舍入误差
- **功能:** 展示大浮点数运算时的精度损失问题, $2.0e20 + 1.0 - 2.0e20$ 的结果不是1.0
- **演示效果:**

```
1  /* floaterr.c--demonstrates round-off error */
2  #include <stdio.h>
3  int main(void)
4  {
5      float a,b;
6
7      b = 2.0e20 + 1.0;
8      a = b - 2.0e20;
9      printf("%f\n", a);
10
11     return 0;
12 }
13
```

[Done] exited with code=0 in 0.303 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\" && gcc charcode.c -o charcode && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\"charcode

[Done] exited with code=1 in 25.087 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\" && gcc escape.c -o escape && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\"escape

[Done] exited with code=1 in 1.797 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\" && gcc floaterr.c -o floaterr && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\"floaterr

4008175468544.000000

[Done] exited with code=0 in 0.307 seconds

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platinum.c

- 作用: 计算用户体重对应的铂金价值
- 功能: 读取用户体重（磅），计算对应的铂金价值（假设铂金价格为\$1700/盎司）
- 演示效果:

```
badcount.c  escape.c  platinum.c X
C platinum.c
4 {
5     float weight; /* user weight */
6     float value; /* platinum equivalent */
7
8     printf("Are you worth your weight in platinum?\n");
9     printf("Let's check it out.\n");
10    printf("Please enter your weight in pounds: ");
11
12    /* get input from the user */
13    scanf("%f", &weight);
14    /* assume platinum is $1700 per ounce */
15    /* 14.5833 converts pounds avd. to ounces troy */
16    value = 1700.0 * weight * 14.5833;
17    printf("Your weight in platinum is worth $%.2f.\n", value);
18    printf("You are easily worth that! If platinum prices drop,\n");
19    printf("eat more to maintain your value.\n");
20
21    return 0;
22 }
23

问题 输出 调试控制台 终端 窗口 SPELL CHECKER 串行监视器
+ v [x] pwh [ ] 窗 ... | [ ]

PS D:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03> .\charcode.exe
Please enter a character.
c
The code for c is 99.
PS D:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03> .\escape.c
PS D:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03> ^C
PS D:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03> .\escape.exe
Enter your desired monthly salary: $9999999
Gee! $9999999.00 a month is $119999988.00 a year.
PS D:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03> .\platinum.c
PS D:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03> ^C
PS D:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03> .\platinum.exe
Are you worth your weight in platinum?
Let's check it out.
Please enter your weight in pounds: 2000
Your weight in platinum is worth $49583220.00.
You are easily worth that! If platinum prices drop,
eat more to maintain your value.
PS D:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03> |
```

print1.c

- **作用:** 演示printf函数的正确和错误用法
- **功能:** 展示printf参数匹配的正确用法和忘记参数的错误用法
- **语义错误:**
第11行: `printf("%d minus %d is %d\n", ten);` - 格式字符串需要3个参数, 但只提供了1个
- **演示效果:**


```
1  /* print1.c-displays some properties of printf() */
2  #include <stdio.h>
3  int main(void)
4  {
5      int ten = 10;
6      int two = 2;
7
8      printf("Doing it right: ");
9      printf("%d minus %d is %d\n", ten, 2, ten - two );
10     printf("Doing it wrong: ");
11     printf("%d minus %d is %d\n", ten ); // forgot 2 arguments
12
13     return 0;
14 }
15
```

问题 输出 调试控制台 终端 窗口 SPELL CHECKER 串行监视器 筛选器 Code

Doing it wrong: 10 minus 2 is 0

[Done] exited with code=0 in 0.295 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\" && gcc print1.c -o print1 && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\print1

Doing it wrong: 10 minus 6422356 is 4200939

Doing it right: 10 minus 2 is 8

[Done] exited with code=0 in 0.308 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\" && gcc print1.c -o print1 && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\print1

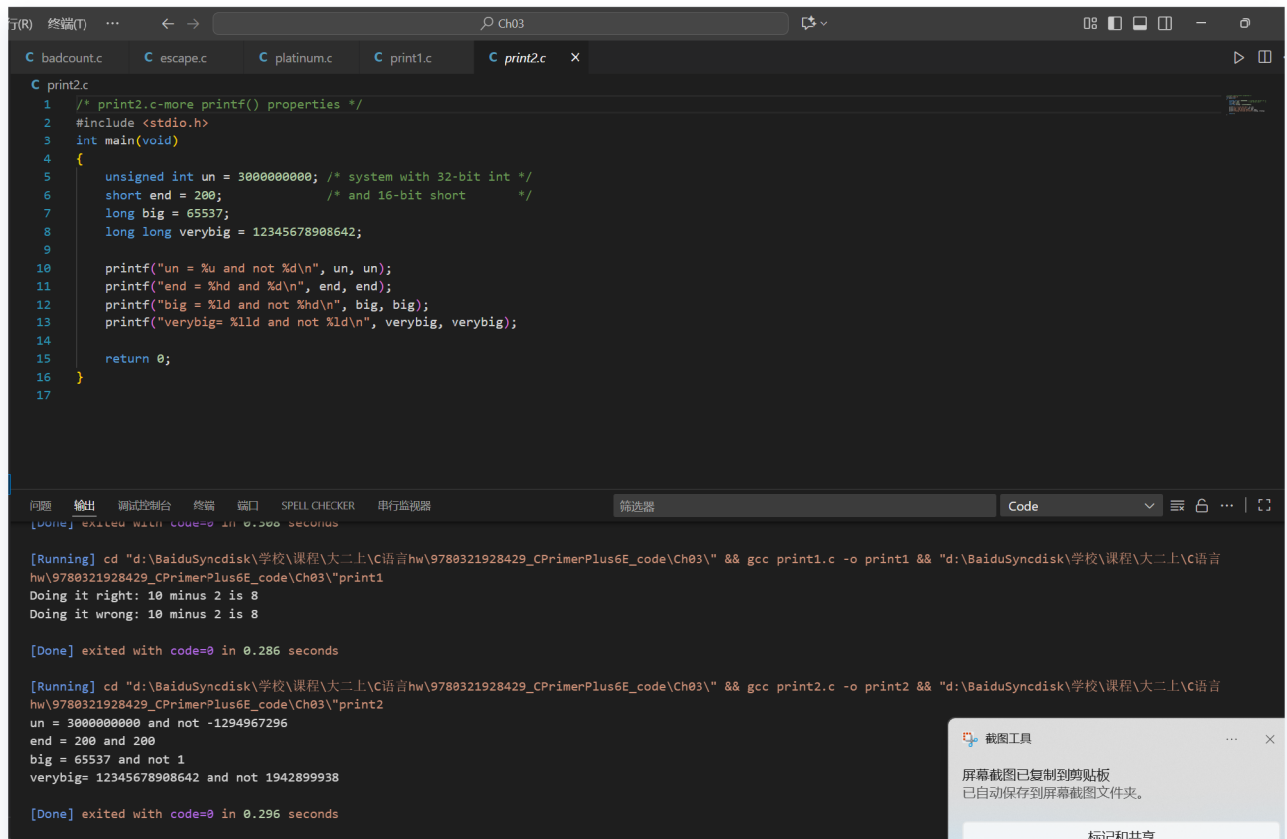
Doing it right: 10 minus 2 is 8

Doing it wrong: 10 minus 2 is 8

[Done] exited with code=0 in 0.286 seconds

print2.c

- **作用:** 演示不同整数类型的输出格式
- **功能:** 展示unsigned int、short、long、long long等类型的正确格式化输出
- **演示效果:**



```
1 /* print2.c-more printf() properties */
2 #include <stdio.h>
3 int main(void)
4 {
5     unsigned int un = 3000000000; /* system with 32-bit int */
6     short end = 200; /* and 16-bit short */
7     long big = 65537;
8     long long verybig = 12345678908642;
9
10    printf("un = %u and not %d\n", un, un);
11    printf("end = %hd and %d\n", end, end);
12    printf("big = %ld and not %hd\n", big, big);
13    printf("verybig= %lld and not %ld\n", verybig, verybig);
14
15    return 0;
16 }
17
```

[Done] exited with code=0 in 0.286 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\" && gcc print1.c -o print1 && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\print1

Doing it right: 10 minus 2 is 8

Doing it wrong: 10 minus 2 is 8

[Done] exited with code=0 in 0.286 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\" && gcc print2.c -o print2 && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\print2

un = 3000000000 and not -1294967296

end = 200 and 200

big = 65537 and not 1

verybig= 12345678908642 and not 1942899938

[Done] exited with code=0 in 0.296 seconds

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showf_pt.c

- **作用:** 演示浮点数的不同表示方式
- **功能:** 展示浮点数的标准格式、科学计数法格式和十六进制格式输出
- **演示效果:**

- [illegible]

- **作用:** 显示各种数据类型的大小
- **功能:** 使用sizeof运算符显示int、char、long、double等数据类型在系统中的字节大小
- **演示效果:**

运行(R) 终端(T) ... ← → Ch03

C badcount.c C escape.c C platinum.c C print1.c C showf_pt.c C typesize.c X

```
C typesize.c
1  /* typesize.c -- prints out type sizes */
2  #include <stdio.h>
3  int main(void)
4  {
5      /* c99 provides a %zd specifier for sizes */
6      printf("Type int has a size of %zd bytes.\n", sizeof(int));
7      printf("Type char has a size of %zd bytes.\n", sizeof(char));
8      printf("Type long has a size of %zd bytes.\n", sizeof(long));
9      printf("Type long long has a size of %zd bytes.\n",
10             sizeof(long long));
11     printf("Type double has a size of %zd bytes.\n",
12            sizeof(double));
13     printf("Type long double has a size of %zd bytes.\n",
14            sizeof(long double));
15     return 0;
16 }
17
```

问题 输出 调试控制台 终端 窗口 SPELL CHECKER 串行监视器 筛选器 Code

```
[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\" && gcc typesize.c -o typesize && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\"typesize
2147483647 -2147483648 -2147483647
4294967295 0 1

[Done] exited with code=0 in 0.297 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\" && gcc typesize.c -o typesize && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch03\"typesize
Type int has a size of 4 bytes.
Type char has a size of 1 bytes.
Type long has a size of 4 bytes.
Type long long has a size of 8 bytes.
Type double has a size of 8 bytes.
Type long double has a size of 12 bytes.

[Done] exited with code=0 in 0.281 seconds
```